(12 points) What will be the values of the specified registers and flags after the execution of the following instructions?

1)

MOV BH, 38H
ADD BH, 2FH

BH = (67)16

CF = 0
SF = 0
PF = 0

AF = 1
ZF = 0
OF = 0

2)

MOV BX, 34F5H
MOV AX, 95EBH
ADD AX, BX

AX = (CAE0)16
BX = (34F5)16

CF = 0
SF = 1
PF = 0

AF = 1
ZF = 0
OF = 0

```
  111
  0011 1000
+ 0010 1111
------------
  0110 0111
```

```
  11   111   1111   111
  0011 0100 1111   0101
+ 1001 0101 1110   1011
-----------------------
  1100 1010 1110   0000
  C    A    E      0
```

(12 points) Given the following initial register values:

- SP = 123C
- AX = 24B4
- BX = 32F4
- DI = 85D2
- DX = 5F93

Complete the provided stack diagram by performing the following tasks:

1. Illustrate the stack contents after each of the following instructions is executed.
2. Show the updated SP position after each operation and its final value.

Instructions to execute sequentially:

1 POP BX
2 PUSH AX
ADD AX, BX
3 PUSH DX
4 PUSH AX

push SP-2
pop SP+2

0000
FFFF

Ensure clarity in stack changes and register modifications at each step.

	1	2	3	4
				1238 SP → F8
			123A SP → 93	41
			5F	93
		123C SP → B4	B4	5F
SS: 123C → 44		24	24	B4
1D	123E SP → 36	36	36	24
36	15	15	15	36
15				15
AX	24B4	24B4	41F8	41F8	41F8
BX	32F4	1D44	1D44	1D44	1D44
DX	5F93	5F93	5F93	5F93	5F93

SP 1238

C. (2 points) Identify whether the following instructions are legal or illegal. Explain the reason for each illegal instruction.

a. MOV EAX, [var1] legal

illegal instruction.

a. MOV EAX, [var1] legal

b. PUSH AL cannot Push 8bit register only 16bits illegal

c. INC AX, 1 illegal should be INC AX without $\boxed{,1}$

d. MOV [SI], [BX] illegal, cant move from memory to memory

B. (2 points) If CS = 24F6H and IP = 6346H, determine:

Segment offset

CS → code
DS → data
ES → Extra
SS → stack

- The logical address. 24F6:6346 ✓
- The offset address. 6346
- The physical address. 24F60 + 6346 = 2B2A6
- The upper range of the code segment.

24F60 + 0000 = lower: 24F60

24F60 + FFFF = upper: 34F5F

A. (2 points) Assume the following register values (in hexadecimal):
DS =4600, SS=3000, BX=2400, SI=1586, DI=9500, BP=6814, CX=5791, DX= 3299 and AX=75FF.

1. Identify the addressing mode used in each of the following instructions.
2. Calculate the physical address of the source operand for each instruction and determine where it is stored.

- MOV AX, [SI] Indirect

46000 + 1586 = PA: 47586
47587

- MOV AX, [BX+DI+08]
Base-index relative

46000 + 9500 + 08 = 4F508
4F508 + 2400 = 51908

PA: 51908
51909

(5 points) Multiple Choice: Select the correct answer for each of the following questions.

1- What is the content of stack pointer
(a) Address of the top element in the stack
(b) Address of current instruction
(c) Address of next instruction
(d) None of the above

SP →

2- Which of the following instruction is not valid?
(a) Mov ax, bx
(b) Mov ds, 7000h → Mov ax, 7000H Mov ds, ax
(c) Mov ax, 4000h
(d) Push ax

3- Which of the following instructions moves data from memory to the AX register?
(a) MOV [data], AX X
(b) MOV AX, data
(c) MOV AX, [data]
(d) MOV [AX], data X

4- A PUSH instruction will cause the SP to be:
(a) Incremented by 1
(b) Incremented by 2
(c) Decremented by 1
(d) Decremented by 2

SP−2 → Push
SP+2 → POP

5- The Intel Architecture system follows which of the following design?
(a) CISC
(b) RISC
(c) All of the above
(d) None of the above

6- To copy the hexadecimal value A to the BH register you write
(a) MOV BH, AH X
(b) MOV BH, 0AH
(c) MOV AH, BH X
(d) MOV BH, A0H X

7- Which of the following registers keeps track of the instructions stored in memory?
(a) Accumulator
(b) Address Register
(c) Instruction Pointer
(d) Index Register

2^{10} → KB
2^{20} MB
2^{30} GB
2^{40} TB

8- A 24-bit address bus allows access to a memory of capacity
(a) 1 Mb
(b) 2 Mb
(c) 32 Mb
(d) None of the above

2^{24} $\boxed{2^{20}}$ 2^{4}
1MB·16 → 16MB

9- In the 8086, the direction flag (DF) is used to control the direction of which operation?
(a) Arithmetic operations
(b) Memory operations
(c) Interrupt operations
(d) String operations

(b) Memory operations
(c) Interrupt operations
(d) String operations

10- The system bus is made up of
(a) Data bus
(b) Data bus and address bus
(c) Data bus and control bus
(d) Data bus, control bus and address bus

Write an assembly program that adds 2 arrays of 4 bytes and saves the result in a third array in the following manner.

CX=4 CX=4 same CX same size CX=4

Array 1	Array 2	Result
25H	2BH	25H+85H = AA
4FH	C4H	4FH+4FH = 9E
85H	4FH	85H+C4H = 149 CF = 1
1FH	85H	1FH+2BH = 4A

SI→ 0 1 2 3 BX+3 DI→

```
.Model Small
.Stack 64
.Data
Arr1 DB 25H,4FH,85H,1FH
Arr2 DB 2BH,C4H,4FH,85H
Res DB 4Dup ?
.CODE
Main Proc FAR
Mov Ax,@DATA
Mov DS,Ax
Mov CX,04
Mov SI,offset Arr1
```

```
Mov CX, 04
Mov SI, offset Arr1
Mov BX, offset Arr2+3
Mov DI, offset Res
Mov AL, 0

Again: Mov AL, [SI]
ADD AL, [BX]
Mov [DI], AL
Inc SI
Dec BX
Inc DI
loop Again  ->  Dec CX
                JnZ Again

Mov AH, 4CH
INT 21H
MAIN ENDP
END MAIN
```

Write an assembly program that adds 2 arrays of 4 words and saves the result in a third array in the following manner.

w = 2B

SI

Array 1
2512H
4F34H
8556H
1F78H

Bx

Array 2
2B12H
C434H
4F56H
8578H

DI

Result
2512H + 8578H = AA8A
AF34H + 4F56H = FE8A
8556H + C434H = 1498A CF = 1
1F78H + 2B12H = 4A8A

120
2B 1
34 2
C4 3
56 4
4F 5
78 6
85 7

```
.Model Small
.Stack 64
.Data
```

```
Arr1   DW   2512H,4F34H,8556H,1F78H
Arr2   DW   2B12H,C434H,4F56H,8578H
Res    DW   4DUP ? = ?,?,?,?
.CODE
MAIN PROC FAR
MOV AX,@DATA
MOV DS,AX
MOV CX,04
MOV SI,offset Arr1
MOV BX,offset Arr2+6
MOV DI,offset Res
MOV AX,0
Again: MOV AX,[SI]
  ADD AX,[BX]
  MOV [DI],AX
  ADD SI,2
  SUB BX,2
  ADD DI,2
  loop Again
MOV AH,4CH
INT 21H
```

```
MAIN ENDP
ENDMAIN
```

- Write a complete assembly language program that calculates the sum of five temperatures (stored in an array of bytes as shown below) and stores the result in the BL register.

```
.data
  Temp DB 19H, 2DH, 30H, 35H, 25H    ; Array of temperatures
```

1 2 3 4 5

```
.Model Small
.Stack 64
.DATA

Temp DB 19H,2DH,30H,35H,25H
Sum DW ?
.CODE
MAINPROC FAR
Mov Ax,@DATA
Mov DS,Ax
Mov CX,05
Mov SI,offset Temp
Mov BX,0000
Again: ADD BL,[SI]
   JNC X1
   INC BH
```

Code

```
    INC BH
X1: INC SI
    loop Again
    MOV SUM, BX
    MOV AH, 4CH
    INT 21H
    MAIN ENDP
    END MAIN
```

Code